Programming Autonomous Robots
COSC2781 (PG) | COSC2814 (UG) | Semester 1 2026
ROS2 Search & Navigation Challenge (v1.1)

Assessment Type	Group Assessment. Grades for UG and PG are the same in this assessment.
ROS Proficiency Due Date	11.59pm, Sunday 22 March 2026 (Week 3)
~~**Demonstrations**~~	~~Thursday Week 6 Workshop (before final submission)~~
~~**Report Due Date**~~	~~11.59pm, Sunday 19 April 2026 (Week 6)~~
~~**Silence Period**~~	~~From 5.00pm, Friday 17 April 2026 (Week 6)~~
Due Date	11.59pm, Sunday 26 April 2025 (Week 7)
Silence Policy	From 5.00pm, Thursday 23 April 2025 (Week 7)
Demonstrations	Thursday Week 7 Workshop (before final submission)
Weight	25% of the final course mark
Submission	Online via Canvas. Submission instructions are provided on Canvas.

Change Log

v1.1
- Revised due-dates

1 Overview

In this assignment you will become familiar with working with the ROSBot Pro 3.0 (and Pro 2.0), developing autonomous robot software, and using ROS2 Jazzy/Humble. You will need to combine multiple common ROS2 software packages (such as mapping, navigation and vision processing), and you will also need to write your own ROS2 nodes to augment existing ROS2 software, to enable the ROSBot to successfully complete the Search & Navigation Challenge. You may use simulation to initially develop and test your ROS2 software, however, your work will be assessed and evaluated on the physical ROSBot. You will be assessed on your ability to demonstrate a proficiency with ROS2, your proficiency in safely operating the physical ROSBot, your ability to choose and integrate with suitable existing ROS2 autonomous software, your own ROS2 node implementations, and the speed and accuracy at which your software completes the challenge on the real ROSBot.

The Search & Navigation Challenge is inspired by autonomous robotics for Urban Search and Rescue (USAR) and the RoboCup Rescue Robot League. In USAR after a disaster occurs, an autonomous robot can explore an unknown environment, such as a

partially collapsed building, and locate trapped or unconscious human victims of the disaster. The robot builds a map as it explores and places markers in the map of where the victim's locations. This map is given to First Responders who use it to go into the building and rescue the victims.

In this assignment you will enable the ROSBot to:

1. Explore an unknown location
2. Locate a series of pre-defined objects
3. Place the objects in the map
4. Return the robot to it's starting position
5. Plot the path the robot took on the map
6. Complete the challenge as fast and as accurately as possible

Before starting work on your implementation, you will also be required to demonstrate your proficiency in ROS2 on TheConstruct, and your proficiency in safely operating the physical ROSBot with ROS2 software.

1.1 Group Work

You are required to work in groups of 3 where possible, and groups of 2 for odd numbers. No other group sizes are permitted. There may be both UG and PG students in the same group. This is because of:

1. The limited availability of the physical robots.
2. To reduce wear-and-tear and prevent over-use of the robots.
3. Safety, in ensuring appropriate monitoring of the robots during autonomous operation.

Note that there are no specific grades for group work in this assessment. As post-graduate and final year undergraduate students, you are expected to have suitable group work skills. The work of the assessment is distributed between the students with individual contributions. We expect that you are able to communicate openly and effectively with your group members, resolve issues, and function coherently as a team. However, in the event of group-work issues, individual grades may be allocated. Therefore, we advise that your group uses suitable group work tools (such as MS Teams and a Git Repository) to track your individual contributions.

We also recommend that you work closely together on the assignment, developing the code together, so that each member of the team has a good understanding of how all of the parts of your final software system work. This understanding will greatly help you in the project.

1.2 Learning Outcomes

This assessment relates to the following course learning outcomes:

CLO1 Discuss and Critically Analyse and a variety of software architectures and algorithms for solving typical problems in the context of autonomous robot systems; Discuss and Critically Analyse the strengths and limitations of these architectures and algorithms.

CLO2 Discuss and Critically Analyse the challenges of designing and developing software for a variety of robot systems of different complexities, including noise, uncertainty, and computational power.

CLO4 Use pre-existing robot software to solve common problems on simulated and real-world robots; Develop and Implemented new algorithms and software for solving problems on simulated and real-world robots; Integrate this software in the ROS framework.

1.3 Use of Artificial Intelligence (AI) Tools in this Assessment

Perhaps compared to other courses, the use of AI tools and Generative AI in this course is generally quite open. This being noted, it remains the case that ***the majority of your work in this assessment should be your own work***, and not plagiarised from other sources, or sourced from the use of Artificial Intelligence (AI) tools. Therefore, the use AI tools are restricted in certain ways for this assessment task.

In this assessment task, you may use AI tools to support you in developing basic ROS2 boilerplate software, generating ideas, planning, and/or drafting. You may find it helpful to use AI Tools for:

1. Boiler-plate ROS code, such as node templates
2. Boiler-plate ROS package and configuration files
3. Ensuring ROS syntax is correct, such as topic types
4. Debugging code compilation errors

We also suggest reviewing the Course AI Policy on Canvas for hazards and pitfalls in using AI tools in this assessment. Any use of such tools must be acknowledged and referenced. Work that is significantly produced by AI tools, or where AI tools are used to complete this assessment without attribution may result in an allegation of academic misconduct.

For the purposes of openness, transparency, and academic integrity, you must submit a log of your use of generative AI tools during this assessment. This also means you should think about how you track your AI use. Each tool has a way of exporting your interaction history. The below links describe how to export your logs from different AI tools:

- ChatGPT
- Claude
- Gemini

Suspected use of AI tools without a log may result in grade penalises and/or misconduct proceedings.

2 ROS2 Proficiency

Before starting work on your implementation, you are required to demonstrate your proficiency in ROS2 through the TheConstruct modules, and in using ROS2 software to work safely with the ROSBot.

In Weeks 1-3, you will complete modules on TheConstruct. Completion of these modules contributes to your grades in this assessment, to encourage you to get an early start on ROS2. You should complete the below Quizzes on the '*ROS2 Basics in 5 Days (Python)*' TheConstruct module by **11.59pm, Sunday 22 March 2026 (Week 3)**. Note that each unit also has a "knowledge check". Quizzes are different. The quizzes are accessible by selecting the relevant chapter from the notebook.

- Unit 3: Understanding ROS2 Topics (chapter 3.10)
- Unit 4: Understanding ROS2 Services (chapter 4.9)
- Unit 7: Understanding ROS2 Actions (chapter 7.9)

Additionally in Weeks 1-3, as part the workshops you will become familiar with using ROS2 on the ROSBot. Safe use of all equipment is very important, and proficient use of the ROSBot's is necessary for completing this assessment. To ensure both, you must individually demonstrate your proficiency in using the ROSBot, by **11.59pm, Sunday 22 March 2026 (Week 3)**, without assistance from staff or other students. The proficiency exercise is listed on the AIIL Bitbucket repository wiki. You can demonstrate this proficiency to any course staff during Week 2 and 3 classes. Please note that **failure** to complete this proficiency exercises is considered a **safety issue**, and may result in access being removed to the labs, re-completion of inductions, and may prevent meeting mandatory course safety requirements.

3 Search and Navigation Challenge

Your groups must create autonomous robot software in a single ROS2 package that enables the ROSBot to successfully complete a *Search & Navigation Challenge* as fast and accurately as possible. The capability of your ROS2 package will be measured during a timed completion of the challenge in an unknown space.

The Search & Navigation Challenge will be conducted as follows:

1. The ROSBot will be placed into a closed space (such as a room or maze)
   - You will not know the exact location and layout of the space will prior to the challenge.
   - The size of the closed space will be set so that the ROSBot is capable of traversing the space in the set time limit.
   - Fixed static obstacles, clearly visible to the ROSBot laser may be placed within the space.
   - No dynamic obstacles will be present.
2. Five (5) hazard markers will be selected from the available set, and placed around the space.
   - The location of the hazard markers will be at a height clearly visible to the ROSBot camera at a distance of at least 10cm.
   - The hazard markers will be fixed to a flat card.
   - See Section 3.3 for a description of the markers.
3. You will launch the software in your ROS package.
   - The marking script that monitors the required topics (see Section 3.5) will be launched and connect to the ROSBot topics.

- After all software is launched, you may not interact with the ROSBot, terminals, or software, unless otherwise directed.
- While your software is running, it must report it's status via a status topic (see Section 3.5).

4. The ROSBot will be signalled to start using the Start Marker, described in Section 3.4.
5. The ROSBot must explore the closed space, completing the following tasks:
   (a) Create and publish an accurate map of the closed space, see Section 3.5 for topic details.
   (b) Locate all 5 hazard markers.
   (c) Publish the location of all of the hazard markers in map relative coordinates, see Section 3.5 for topic details.
   (d) Publish the path the robot travels in the closed space in map relative coordinates, see Section 3.5 for topic details.
6. Once all hazard markers have been located, or via a manual trigger sent after at most 4 minutes have elapsed, the ROSBot must return to its starting location (at any orientation) by retracing the path it took to explore the environment, see Section 3.9.
7. During the challenge, contingencies may be invoked as described in Section 3.10.
8. The challenge ends if:
   (a) The ROSBot returns to it's original location and stops for at least 5 seconds.
   (b) 7 minutes have elapsed.
   (c) You (the students) or the assessor request the challenge to end.
   (d) The ROSBot touches a wall.
   (e) A safety concern is identified.
   (f) There is external interaction with the ROSBot or software.

Your group will need to write three ROS2 nodes:

1. Navigation logic, described in Section 3.7.
2. Hazard Marker detection and placement, described in Section 3.8.
3. Position Tracking and Return to Home Logic, to track the path of the robot, described in Section 3.9.

Each student in the group is responsible for one node. This doesn't preclude working together, as software integration is needed to support all nodes working together. You may also need to write additional ROS2 nodes to support functionality not covered in the three required nodes.

Groups of 2 should implement nodes #1 and #2 as above. In addition node #2 must include the component of node #3 to track and publish the robot's path in exploration.

3.1 Autonomy

Your ROS2 software for the challenge must be fully autonomous. The only interaction with the software is through the Start Marker, status triggers, and contingencies (see Section 3.10).

3.2 Real-Robot Evaluation

All evaluations of the successful implementation and performance of your ROS2 nodes and software for the challenge, must be conducted on the real ROSBot robots in the AI Innovation Lab (14.11.01) and/or Cyber Security Lab (14.11.38). While you may use simulation (such as on TheConstruct) to initially develop and test your software, the evaluation will only be conducted on the real-robot.

3.3 Hazard Markers

The hazard markers are selected from the symbols used in the RoboCup Rescue Robot League. The markers are available on the Canvas Assignment module. This collection contains more markers than the 5 that will be used during the Search & Navigation Challenge. Your software should be capable of identifying all of the markers, as you will not know prior to the challenge the exact set of markers that will be used.

To ensure the correct identification of the hazard markers, you must use the following IDs when publishing the hazard marker locations.

Hazard Marker	ID
Unknown	0
Explosive	1
Flammable Gas	2
Non-Flammable Gas	3
Dangerous When Wet	4
Flammable Solid	5
Spontaneously Combustible	6
Oxidizer	7
Organic Peroxide	8
Inhalation Hazard	9
Poison	10
Radioactive	11
Corrosive	12

3.4 Start Marker

The start marker is used to indicate to the ROSBot that it should commence the Search & Navigation Challenge. This marker is available on the Canvas Assignment module. Note that the start marker *must not* be published on the hazard markers topic.

3.5 Published topics

You are required to publish the following topics. Output from these topics will be captured during the challenge.

Element	Topic	Message Type
Map	`/map`	`nav_msgs/msg/OccupancyGrid`
Hazard Markers	`/hazards`	`visualization_msgs/msgs/Marker`
Robot Path (Explore)	`/path_explore`	`nav_msgs/msgs/Path`
Robot Path (Return Home)	`/path_return`	`nav_msgs/msgs/Path`
Status	`/snc_status`	`std_msgs/msgs/String`

Take care of these additional notes on the information to publish:

- The ID of the hazard markers should be set according to the table in Section 3.3.
- The hazard markers and robot paths **must** be published in the map frame. This is important. ROS2 allows you to publish these elements in any frame, and if you are viewing these through RViz/Foxglove, the visualisation software will transform these elements into other frames for *visualisation purposes only*. Part of the challenge of this assignment is computing correct frame transforms. Thus the marking collection script will assume a map frame of reference regardless of the published frame.
- The contents of the status message should be a simple string indicating what the robot is attempting to do at any point in time. This is used for both yourselves and the assessor to understand the robot state. Exact message formats are not specified.

3.6 Trigger Topics

To support contingencies in completing part of this assessment, your code may support the following triggers. These triggers will be called by the assessor to invoke a contingency. We have chosen to implement these triggers as topics (publisher/subscriber) model rather than Services for simplicity of implementation.

Element	Topic	Message Type	Purpose
Start Marker	`/trigger_start`	`std_msgs/msgs/Empty`	Contingency if visually recognising the start marker fails.
Teleop	`/trigger_teleop`	`std_msgs/msgs/Empty`	Contingency to teleoperate (manually drive) the robot for exploring the map if the exploration node fails.
Return Home	`/trigger_home`	`std_msgs/msgs/Empty`	Contingency to terminate exploration and have the robot return to home following it's exploration path if this functionality is not automatically triggered.

3.7 ROS2 Node #1: Navigation Logic

You must write your own ROS2 node to enable the navigation of the ROSBot within the closed space, in searching for the Hazard Markers. This logic might include:

- Start navigation when the marker is shown.
- Determining where (and how) to move about the space.
- Stopped (and spinning on the spot) to locate hazard markers.
- Stopping to more accurately place a hazard marker in the map.
- Returning to the starting location.

You will need various elements of the ROS2 Nav2 stack. There is no single elements of the Nav2 stack that does exactly what is needed for the Search & Navigation Challenge. Thus, you will need to devise your *own* algorithm. There are three recommended approaches to begin with:

1. Use the Occupancy grid to look for unknown regions of the map.
2. Use the Explore Lite package.
3. Use the wall-following example from TheConstruct.

Regardless of what method you choose, a valid implementation of this component, **must** show functionality beyond just executing an existing algorithm.

3.8 ROS2 Node #2: Hazard Marker Detection and Placement

You must write your own ROS2 node to determine the map-relative coordinate of each hazard marker in the closed space by using the *kinematics* (that is, *transform tree*) of the ROSBot. The `find_object_2d` package will tell you if a hazard marker has been identified in the ROSBot camera image. However, the packages *does not* give the map-relative coordinates of the hazard marker. In fact, the `find_object_2d` only provides a bounding-box around the location of the detected object in the camera image.

Instead, you will need to write your own computation to use the *transform tree* to deduce suitable map-relative coordinates of the hazard marker. There are three recommended approaches:

- That front-facing IR range sensors. The range sensors measure points directly in front of the ROSbot, but only return a single value. These could be used to compute an estimate of the distance to the wall(s) in front of the ROSBot.
- The laser. The laser provides distances from the robot to where each laser points intersects the environment. You can use this to estimate the distance of the hazard markers from the robot, and hence the compute the desired coordinates via the transform tree.
- The depth image. The depth image provides distances from the robot to where each pixel of the depth image intersects the environment. Similar to the laser to can ne used to determine the desired coordinates.

Regardless of what method you choose, a valid implementation of this component, **must** use some form of secondary sensor input to the RGB camera to compute the hazard marker position.

3.9 ROS2 Node #3: Robot Position Tracking and Return to Home

You must write your own ROS2 node that:

1. Tracks and publishes the path of the robot taken while exploring.
2. Returns to the robot's home (starting) position by following this path in reverse order.
3. Tracks and publishes the path of the robot taken when returning to home.

To track the robot's path, you will need to periodically capture the location of the robot. Recall, that the robot's map coordinate can be determined by using a *kinematic* transform between the robot frame and the map frame. Take care to publish the history of the robot's path on the correct topic, as there are separate topics for exploring and returning to home.

To implement the return-to-home logic, you could consider using:

1. Nav2 Stack waypoint-list following.
2. Nav2 Stack single waypoint target, with your own custom logic for selecting the next waypoint.
3. Your own drive commands, such as PID control to reach each waypoint in sequence.

For any of these approaches, you will need to carefully consider the detail of the history of the path the robot took in exploring. Too many points in this history will negatively impact the return-to-home logic, but too few points will fail to follow the original path.

Regardless of what method you choose, a valid implementation of this component, **must** attempt to follow the path that was explored to some extent. Simply navigating to the starting position is insufficient.

3.10 Contingencies

In the event of issues with some of the ROS2 nodes, the following contingencies are put in place so that functionality of other nodes can be demonstrated. Invoking a contingency may impact your grade and final competitive ranking. The contingencies use the Trigger Topics described in Section 3.6. The contingencies are:

1. If the start marker cannot be visually identified, then the exploration phase of the robot can be commenced by using the 'Start Marker' trigger.
2. If the exploration ROS2 node does not satisfactorily explore the environment, then the exploration phase may be done via teleoperation (manually driving the ROSBot). This mode will be communicated to your code by the 'Teleop' trigger, after which your group may launch a teleop ROS2 node, such as keyboard-teleop, or Foxglove Web-UI teleop.
3. If the return-to-home logic is not automatically invoked, then it may be manually invoked using the 'Return Home' trigger.

3.11 Collaboration and Maintenance

We have regularly noted that this course only functions if all students (and staff) collaborative work including maintaining the equipment and AIIL space. Documentation and good maintenance can often be a thankless job, but is critical for all students successfully completing the assignment. Additionally, ROS2 Jazzy is new for this semester, so learning ROS2 is a process for everyone.

Therefore, during the completion of this assessment you are asked to be a "good course citizen". This includes:

1. Generating and maintaining course documentation including (but not limited to) working in ROS2 Jazzy, the use of the ROSBot, the use of TheConstruct, and the status of each ROSBot. Documentation is maintained on BitBucket, and Canvas.
2. Providing collaborative help and assistance to other students in the course (without directly enabling plagiarism) for shared information. This help should be recorded on the course MS Teams channels.
3. Actively ensuring robot maintenance including regular charging of the ROSBots, general care of the ROSBots, and keeping the AIIL space clean and tidy.
4. Timely reporting of incidents or faults in the AIIL including the general space and any AIIL equipment, ROSBots included.

4 Challenge Demonstrations

> **!** The demonstration will happen *before* you make your final submission.

The primary assessment of the Search & Navigation Challenge is in a common evaluation. This will be a live demonstration of the challenge during **Thursday Week 7 Workshop (before final submission)**. The demonstration will happen *before* you make your final submission so that you have time *after* the demonstration to complete your final report.

A schedule of all groups will be provided closer to the demonstrations. You should not assume that you will have a particular robot to use.

The Search & Navigation Challenge will be conducted as described in Section 3, with the additional notes:

- To facilitate time constraints of the expected number of groups, the demonstrations may commence earlier and run later than the scheduled times of the Thursday workshop.
- Each group will have up to 2 attempts at the Search & Navigation Challenge, subject to time constraints. The best attempt will be used for assessment and final rankings.
- If a group is unable to successfully complete a significant portion of the challenge they may be granted additional attempts, however, these will not count toward the final rankings.
- The closed space may be modified between each attempt.
- The hazard makers and locations may be changed between each attempt.

4.1 Final Rankings

A small portion of the grade is for comparing the speed and accuracy of all groups. Each successful attempt will be considered across categories of:

1. Speed.
2. Accuracy of the Map.
3. Accuracy of the placed Hazard Markers.
4. Accuracy of the robot paths.
5. Accuracy of the return-to-home path.
6. Straight line distance of the robot's finishing position to the start location, without intersecting walls.
7. Number of invoked contingencies.

To count toward the final ranking the attempt must demonstrate a reasonable completion for one of the three required nodes. That is, the node should at a minimum publish reasonable information (that is, not testing/fake data or widely inaccurate data) for nodes that publish data, and take reasonable movement for nodes that move the ROSBot. Except in the cases of individual grading, an attempt must:

- Be a reasonable attempt for at least one node to be eligible for a rank.
- Be a reasonable attempt for at least two nodes to be eligible for a rank in the top-66%.
- Be a reasonable attempt for all three nodes to be eligible for a rank in the top-33%.

These ranks are to inform and distinguish between the top performing (and HD) assessments. Final ranks may inform project allocations for high-demand and limited use robots, where only a limited number of groups may work with a particular platform.

5 Evaluation Report

The majority of grades, described in Section 7, are for the quality of your implementation. The challenge demonstration provides a fair and common evaluation for everyone, but it is not the only opportunity to evaluate your work. This is especially the case if your challenge demonstration performance isn't strong. Thus, your group must submit and present an evaluation report, due by **11.59pm, Sunday 26 April 2025 (Week 7)**, after the completion of the challenge demonstration.

The purpose of your evaluation report is to show the qualify of your implementation, and will help inform your grades for the implementation components of the rubric. This means that irrespective of your challenge demonstration, you can receive high marks for the implementation. The effort in collecting good evidence to complete the evaluation report is *as important as* your code. A small number of marks is given for compiling the report itself, as this takes time and effort, but the report *strongly informs* the implementation rubric components and significant contributes to those marks. Of course, a good showing in the challenge demonstration will provide excellent evidence for your evaluation report.

Your evaluation report must be a PDF of no more than 5 pages, and should be accompanied by video evidence. Video evidence should be uploaded to MS Teams before the due date, in a predetermined location listed on canvas. Your evaluation report and accompanying evidence should provide:

- A brief summary of the ROS2 node(s) that you wrote, describing their functionality along with a brief technical summary of your ROS2 package and software including:
  - All ROS2 dependent libraries packages that are used in your software.
  - Description of launch files, custom configuration files, and training data of ML algorithms.
- An analysis and evaluation of your software. This should include:
  - An evaluation of the performance of your software, both during the challenge demonstrations, and from your own independent testing.
  - Analysis and explanation for the strengths of your software.
  - Analysis and explanation of the limitations of your software, which may prevent your software completing some aspect of the Search & Navigation Challenge.
  - Clear and accurate evidence that supports your analysis. That is, any claims in your analysis must be supported by evidence. Generalisations, invalid statements, or unsupported claims will be unlikely to received marks.
- Citations to all reference material, including a brief summary of your use of AI Tools in this assessment.

In collecting your evidence, we recommend that you think about:

- Collecting evidence all the way through this assessment, not just in a rush towards the end.

- Evidence that you are outputting the required and correct data on the correct topics.
- How you can collect evidence to prove that it is your work being evaluated.
- How you can collect evidence that can be easily seen on video. For example, long and fast-scrolling terminal output is not good evidence, and we won't be expected to 'interpret' your terminal output.

6 Submission

Follow the instructions **on Canvas** to complete your submission for the Assignment.

Assessment declaration: When you submit work electronically, you agree to the assessment declaration.

6.1 Silence Period

A silence policy will take effect from **5.00pm, Thursday 23 April 2025 (Week 7)**. This means no questions about this assignment will be answered, whether they are asked on the MS Teams, by email, or in person. Make sure you ask your questions with plenty of time for them to be answered. There may not be time to answer questions which are asked too close to the silence deadline.

6.2 Late Submissions & Extensions

A penalty of 10% per day is applied the grade for the assessment excluding the competitive rank, for late submissions of the evaluation report (that is, after 11.59pm, Sunday 26 April 2025 (Week 7)) up to 5 days, after which you will lose ALL the assignment marks. Extensions will be given only in exceptional cases. Refer to RMIT's Special Consideration process for more information.

Take special note that you are expected to spend significant time on-campus working with the real-robots during this assessment. Being unable to attend on-campus, unless accompanied by evidenced documentation sufficient under the Special Consideration policy, is generally not grounds for an extension or an assessment adjustment.

Special Considerations given after grades have been released and/or solutions have been discussed *will automatically result in an equivalent assessment*, which may take a different form, assessing the same knowledge and skills of the assignment.

7 Marking guidelines

The marks are divided into three categories:

- ROS2 Proficiency (4/25)
- Implementation (15/25), broken down into 3 sub-sections:
  - Implementation of the ROS2 node #1 for Navigation Logic (5/25)
  - Implementation of the ROS2 node #2 for Hazard Marker Detection and Placement (5/25)
  - Implementation of the ROS2 node #3 for Robot Position Tracking and Return to Home (5/25)
- Evaluation Report (2/25)
- Final Ranking (4/25)

The detailed breakdown of this marking guidelines is provided on the rubric linked on Canvas.

All students in the group will receive the same grade, with the exception of grades for completing the ROS2 proficiency exercises. While we encourage distribution of implementing the assessment among the group members, we expect all group members to work together collaboratively in completing this assessment and contribute to solving problems in all elements of this assessment. In the event of group-issues, your assessor may decide to award individual grades.

For groups of 2, the grades for implementation are instead distributed as:

- Implementation of the ROS2 node #1 for Navigation Logic (7.5/25)
- Implementation of the ROS2 node #2 for Hazard Marker Detection and Placement (7.5/25)

8 Academic integrity and plagiarism (standard warning)

Academic integrity is about the honest presentation of your academic work. It means acknowledging the work of others while developing your own insights, knowledge and ideas. You should take extreme care that you have:

- Acknowledged words, data, diagrams, models, frameworks and/or ideas of others you have quoted (i.e. directly copied), summarised, paraphrased, discussed or mentioned in your assessment through the appropriate referencing methods
- Provided a reference list of the publication details so your reader can locate the source if necessary. This includes material taken from Internet sites. If you do not acknowledge the sources of your material, you may be accused of plagiarism because you have passed off the work and ideas of another person without appropriate referencing, as if they were your own.

RMIT University treats plagiarism as a very serious offence constituting misconduct. Plagiarism covers a variety of inappropriate behaviours, including:

- Failure to properly document a source
- Copyright material from the internet or databases
- Collusion between students

For further information on our policies and procedures, please refer to the RMIT Academic Integrity Website.

The penalty for plagiarised assignments include zero marks for that assignment, or failure for this course. Please keep in mind that RMIT University uses plagiarism detection software.